

Junyoung Ki

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EDUCATION

Keimyung university |

Daegu, South Korea

B.S. in Computer Science | GPA: 3.85/4.5

May 2026

Relevant Courses: Computer Graphics (CS250), Data Structures, Linear Algebra, 3D Math for Games, Real-Time Interactive Simulation.

TECHNICAL SKILLS

Programming Languages: C++, C, GLSL

Graphics & Tools: OpenGL 4.6, WebGL 2.0, Git 2.48, GitHub, Visual Studio(2022)

Interpersonal Skills: Solution-Oriented, Detail-Oriented, Adaptable, Passionate about Performance Optimization.

PROJECTS

Keep Breathing | Tech Lead

Dec/2024

- 2D side-view simulation game with visual studio (C++)
- Engineered core gameplay systems including grid-based movement algorithms, AI behavior trees, and combat resolution mechanics.
- Awarded the Most Creative Award by Appian and Best Innovation in Collaboration Award by Red Ventures

Dragonic Tactics | Tech Lead

May/2026

- 2D Turn-based Tactical RPG game collaborating with a cross-functional team of 5 members a custom C++ engine
- Engineered intricate simulation details, including an oxygen capacity system, procedural marine life behaviors, and a specialized 2D perspective view.
- Collaborated closely with the design lead, contributing to game mechanics and technical feasibility to bridge the gap between creative vision and implementation.
- Directed the technical roadmap, ensuring that complex systems like marine simulations remained optimized and scalable.

Custom Multithreaded Rendering Engine | Project manager

Feb/2026

- Architected the engine utilizing Data-Oriented Design (DoD) and Structure of Arrays (SoA) to maximize CPU cache coherency and data throughput.
- Engineered a custom job system from scratch to achieve efficient multithreaded rendering, significantly reducing CPU overhead by distributing tasks across multiple CPU cores.
- Developed custom memory allocators to minimize dynamic allocation overhead, reduce memory fragmentation, and improve memory access speeds.
- Integrated ImGui to build a custom real-time profiling dashboard, enabling on-the-fly monitoring of thread execution times, task distribution, and memory metrics.

LEADERSHIP EXPERIENCE

DigiPen Institute of Technology | Technical Director / Lead Programmer | Daegu, South Korea

- Established and enforced C++ coding standards and best practices for the engineering team.
- Designed the software architecture and delegated specific programming tasks based on team members' strengths.
- Mentored junior programmers on Git workflows, debugging techniques, and Data-Oriented Design principles.